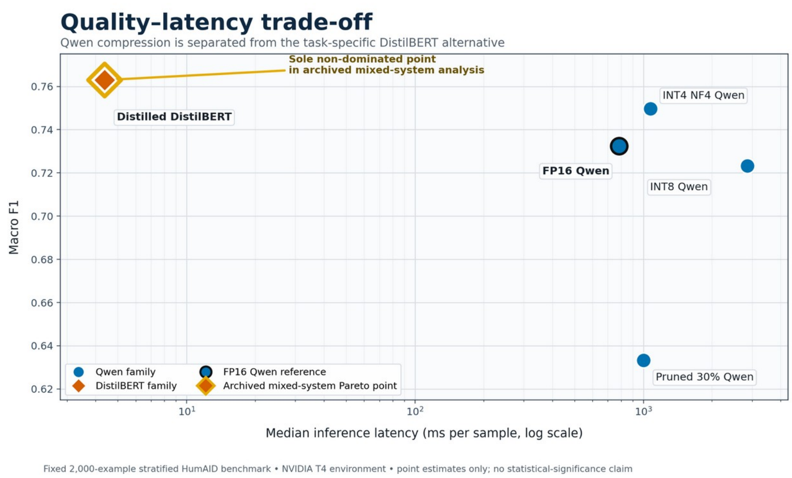
Green AI Compression for Disaster-Tweet Classification

Qwen quantization and pruning, separated from task-specific DistilBERT deployment

0.7497	0.7630	4.38 ms	66.96M
INT4 Qwen macro F1	Distilled student macro F1	Student median latency	Student parameters



Macro F1 vs median latency; log x-axis. Fixed 2,000-example HumAID benchmark, NVIDIA T4.

What the study found

Architecture-preserving Qwen: INT4 NF4 achieved 0.7565 accuracy and 0.7497 macro F1 while reducing recorded peak allocated GPU memory by 65.77% relative to FP16. It was slower and used more estimated energy than FP16 on this T4 setup.

Pruning boundary: 30% one-shot unstructured pruning fell to 0.6395 accuracy; 50% produced 99.6% unmatched outputs, and 70% produced 100% unmatched outputs.

Task-specific alternative: distilled DistilBERT reached 0.7735 accuracy and 0.7630 macro F1. Its approximately 256.13 MB checkpoint used 0.2681 GB peak allocated GPU memory and achieved about 227.36 samples/s formal throughput. It is not a compressed Qwen checkpoint.

Interpretation and limits

The distilled student is the sole non-dominated fully measured point in the archived mixed-system analysis. Within Qwen, FP16 and INT4 retain different advantages. Rare-class results are mixed; missing/found support is only 9. CodeCarbon energy/emissions are estimated, not direct power measurements. Results are point estimates from one dataset and environment.

Authors: Mayuresh Sharma and Aditya Chauhan
GGSIPU, New Delhi | B.Tech CSE, 3rd year
mayureshsharma10406@gmail.com | aditya29ch@gmail.com
Repository: https://github.com/Lucky10406/LLM-compression-study